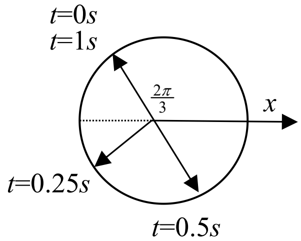
练习九　机械振动（一）

1．$1s$，$\frac{2}{3}\pi$，$\frac{14}{3}\pi$，$5s$

2．见右图



3．（3）

4．（3）

5．（1）$x_0 = 0.5\cos\left(-\frac{\pi}{2}\right) = 0m$，$v_0 = -2.5\sin\left(-\frac{\pi}{2}\right) = 2.5\ m\cdot s^{-1}$

（2）在振动方程中，令$t = \frac{4}{3}\pi$得，$x = 0.5\cos\left(\frac{37}{6}\pi\right) = 0.433m$

又 $v = -2.5\sin\left(\frac{37}{6}\pi\right) = -1.25m/s$，$a = -12.5\cos\left(\frac{37}{6}\pi\right) = -10.8m/s^2$

（3）由$x = 0.5\cos\left(5t - \frac{\pi}{2}\right) = \pm 0.25$，$v = -2.5\sin\left(5t - \frac{\pi}{2}\right) > 0$

得$(5t - 2\pi) = \frac{4\pi}{3}$，$\frac{5\pi}{3}$，$\therefore\ v = -2.5\sin\left(5t - \frac{\pi}{2}\right) = 2.165m/s$

$a = -12.5\cos\left(5t - \frac{\pi}{2}\right) = \mp 6.25m/s^2$，$F = ma = 0.04a = \mp 0.25N$

6.（1）$A = 0.04m$，$\omega = \frac{2\pi}{T} = 2\pi$，由$x_0 = \frac{A}{2}$，$v_0 > 0$，得$\varphi = -\frac{\pi}{3}$　$\therefore\ x = 0.04\cos\left(2\pi t - \frac{\pi}{3}\right)m$

（2）$$\varphi_i = \left(2\pi t_i - \frac{\pi}{3}\right) = \begin{cases} a\text{点}: x_a = A, v_a = 0 \rightarrow \varphi_a = 0,\ t_a = \frac{1}{6}s \\ b\text{点}: x_b = A/2,\ v_b < 0 \rightarrow \varphi_b = \frac{\pi}{3},\ t_b = \frac{1}{3}s \\ c\text{点}: x_c = -A,\ v_c = 0 \rightarrow \varphi_c = \pi,\ t_c = \frac{2}{3}s \end{cases}$$